

OmniDex: Scaling Dexterous Hand Grasping to Diverse Cluttered Scenes

Anonymous Author(s)

Affiliation

Address

email

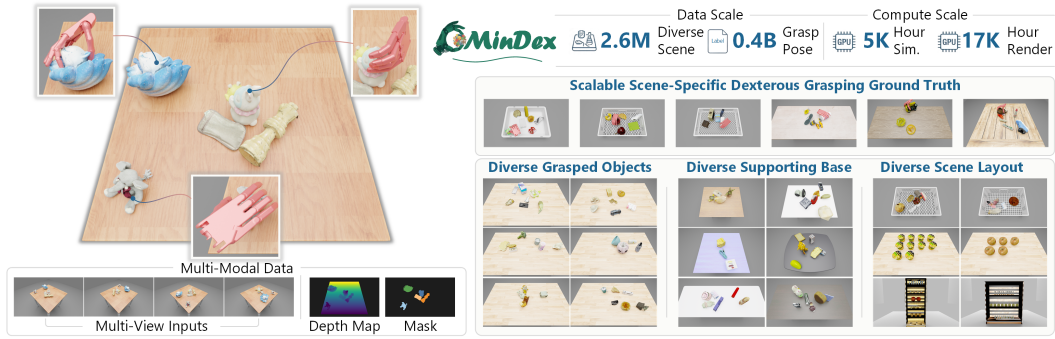


Figure 1: **Overview of OmniDex**, the largest benchmark for dexterous grasping in cluttered scenes to date. **Left:** Multi-view RGB-D and instruction masks paired with scene-specific grasp poses. **Right:** A scalable seed-and-filter strategy enables unprecedented scale (2.6M scenes, 0.4B grasp ground truths), encompassing diverse objects, supporting bases, and realistic layouts.

Abstract

Dexterous grasping is the foundational primitive in embodied AI, demanding massive data to train robust models. As real-world data collection is expensive, simulation has become the mainstream paradigm. Yet, while cluttered scenes best reflect real-world applications, learning to grasp within them is bottlenecked by a critical scarcity of large-scale data. To resolve this, we curate high-quality 3D objects and supporting bases, proposing a scalable seed-and-filter strategy that bypasses sluggish scene-level optimization. This yields an unprecedented benchmark comprising over 2.6 million scenes and 0.4B scene-specific grasp ground truths, featuring diverse realistic layouts paired with rich semantic and geometric observations. Furthermore, we introduce the OmniDex model to overcome the grasp multimodality and last-millimeter precision errors plaguing current generative models. By coupling Soft Winner-Takes-All learning with human-inspired physical constraints during training, and utilizing physics-driven ranking, our approach achieves robust dexterous grasping without the latency of post-optimization. Experimental results show that OmniDex model achieves state-of-the-art performance and strong generalization across diverse scenes, views, and unseen objects.

1 Introduction

Dexterous hands mark a critical milestone in embodied artificial intelligence (AI) Duan et al. [2022]. Unlike parallel-jaw grippers Fang et al. [2020, 2023], they offer the kinematic flexibility and fine-

Table 1: Quantitative comparison of simulation-based dexterous grasping benchmarks. Our dataset features the large scale of scenes and objects, emphasizing diverse layouts in clutter.

Method	# Scenes	Setting	# Objects	Workspace	Modality
Dex1B Ye et al. [2025]	-	Single	6K	-	Depth
AffordDexGrasp Wei et al. [2025]	2K	Single	1K	Table	RGB-D
DDGC Lundell et al. [2021]	0.4K	Cluttered	0.3K	Table	RGB-D
DexGraspNet 2.0 Zhang et al. [2024]	8K	Cluttered	1K	Table	Depth
ClutterDexGrasp Chen et al. [2025b]	1K	Cluttered	2K	Table	Depth
Ours	2.6M	Cluttered	6K	Table, Box, Shelf	RGB-D

grained manipulation essential for complex, human-centric tasks. At the core of this capability lies grasping—the foundational primitive upon which all advanced interactions are built. For an intelligent agent, predicting a valid grasp pose acts as the "key frame" that dictates the success of all subsequent tasks. However, training grasp pose generation models is severely bottlenecked by a profound scarcity of high-quality data. Since collecting high degree-of-freedom (DoF) grasp poses in the real world is expensive and unscalable Liu et al. [2024], simulation-based data synthesis Ye et al. [2025], Chen et al. [2025a] has emerged as the mainstream paradigm. To drive dexterous grasping task toward greater robustness and real-world applicability, this paper introduces a large-scale simulation benchmark targeting cluttered scenes, coupled with a novel learning framework.

Real-world dexterous grasping is inherently messy and cluttered. Yet, the standard setting in prior dexterous grasping literature oversimplifies the problem by isolating a single object in floating space or on a clear tabletop Wang et al. [2023], Xu et al. [2023]. Recently, pioneering efforts have attempted to tackle cluttered scenes Zhang et al. [2024, 2025]; however, robust performance in complex real-world settings remains an open problem due to three underlying challenges. First, current approaches rely on depth maps to form pseudo point clouds as their primary perceptual input. This forces models to focus on geometric shapes, leading to a critical loss of semantic awareness. Consequently, agents struggle with target object localization in cluttered scenes (e.g., distinguishing geometrically identical Coke and Sprite cans), since their geometries are nearly identical. Second, existing benchmark scales are critically constrained by computationally expensive data generation pipelines. Scene-level optimization for grasp ground truths Zhang et al. [2024] incurs prohibitive time costs, limiting existing cluttered-scene benchmarks to only 0.4K–8K scenes (e.g., DexGraspNet 2.0: 8K; Tab. 1). Yet, massive scale is indispensable. Unlike single-object floating setups, cluttered scenes feature complex spatial dynamics among object. Thus, models require a large-scale dataset to effectively learn these spatial relationships and master scene-specific grasping, such as avoiding adjacent items or table penetration. Finally, prior works neglect diverse supporting bases and spatial layouts, compromising robustness and real-world generalization.

Beyond the data bottleneck, existing dexterous grasping generation models face critical algorithmic hurdles when mapping perceptual inputs to high-DoF kinematic space. Dexterous grasping is a multimodal problem, as an object affords numerous viable grasp poses. To capture this complex distribution, data-driven approaches rely heavily on intermediate representations, such as affordance maps Zhang et al. [2024], He et al. [2025], rendering the generation pipeline inherently non-end-to-end. However, pure data-driven methods struggle with critical last-millimeter precision errors. A mere millimeter of deviation translates directly to catastrophic contact failures in simulation. To compensate for this inherent inaccuracy, current methods resort to costly post-optimization or test-time fine-tuning Wei et al. [2025], Zhang et al. [2025], severely compromising inference efficiency.

To address these data and algorithmic bottlenecks, we introduce a holistic paradigm comprising a large-scale cluttered scene benchmark and a novel end-to-end generation framework. To bypass the prohibitive computational cost of scene-level optimization, our OmniDex benchmark employs a scalable seed-and-filter strategy. We first establish a reusable library of free-space seed grasps for single objects, then filter them through collision constraints in cluttered scenes within gravity-stabilized simulations. Leveraging high-quality 3D object assets and high-fidelity rendering, this pipeline yields multi-modal RGB-D observations from multiple fixed camera views, aligned with scene-aware grasp ground truths. Ultimately, our OmniDex benchmark delivers unprecedented scale: over 2.6 million scenes and 0.4B scene-specific grasp pose labels, covering diverse objects, supporting bases, and realistic cluttered layouts. We hope this benchmark will spur a new wave of research in cluttered scenes dexterous grasping. Empowered by this benchmark, OmniDex model fundamentally avoids the pitfalls of non-end-to-end intermediate representations and sluggish test-time fine-tuning. Instead, we internalize physical dynamics directly into training. By coupling a Soft Winner-Takes-All

learning to capture grasp multimodality with a human-inspired physics-constrained loss—which enforces direct hand-object interaction—we effectively bridge the critical last-millimeter precision gap to ensure close and stable grasping. During inference, the optimal grasp pose is rapidly identified via a physical ranking mechanism, achieving both exceptional fidelity and real-time efficiency.

- We introduce a massive cluttered-scene OmniDex benchmark via a scalable seed-and-filter paradigm, yielding an unprecedented dataset of over 2.6 million realistic scenes and 0.4B scene-specific validated grasp ground truths, featuring diverse supporting bases and rich multi-modal observations.

- We propose OmniDex model, a novel end-to-end grasp generation framework that couples Soft Winner-Takes-All learning with human-inspired physical constraints to capture grasp multimodality and bridge the last-millimeter precision gap, with physics-driven ranking for robust inference.

- Experimental results show that OmniDex model achieves state-of-the-art performance on the OmniDex benchmark. OmniDex model demonstrates strong generalization across diverse scene subsets, supporting bases, input views, and unseen objects, highlighting its robustness in realistic cluttered-scene grasping.

2 Related Works

2.1 Dexterous Grasping Datasets and Benchmarks

Real-world dexterous grasping datasets largely rely on human teleoperation Liu et al. [2024], Wang et al. [2025], Xu et al. [2025], Hang et al. [2024]. While capturing human-like priors, this labor-intensive paradigm limits dataset scale and object diversity. Consequently, the community shifted to simulation-based synthesis for isolated objects. Pioneering works Wang et al. [2023], Zhong et al. [2025], Chen et al. [2025a], Wan et al. [2023] leveraged optimization and physical priors to generate massive single-object grasps, culminating in billion-scale datasets like Dex1B Ye et al. [2025]. Despite this impressive scale, these single-object benchmarks fail to model the spatial dynamics of real-world clutter. While recent efforts have introduced cluttered-scene benchmarks Zhang et al. [2024], Chen et al. [2025b] to bridge this reality gap, they remain bottlenecked by costly per-scene optimization. In stark contrast, our efficient seed-and-filter strategy entirely bypasses this sluggish optimization, yielding an unprecedented 2.6 million complex scenes and 0.4B physically validated ground truth, thereby establishing a new benchmark for diversity and scale in cluttered grasping.

2.2 Dexterous Grasp Generation Methods

Learning-based generative models promise real-time dexterous grasp generation, yet mapping visual inputs to high-DoF kinematics is fundamentally plagued by grasp multimodality. To capture this one-to-many manifold, data-driven methods typically rely on heuristic proxies—such as contact maps Xu et al. [2023], Zhang et al. [2025] or affordance fields Zhang et al. [2024], Qin et al. [2023]—rendering the pipeline inherently non-end-to-end. Furthermore, these models struggle with the critical last-millimeter precision, where minute deviations cause catastrophic contact failures. To mitigate this, recent frameworks Wei et al. [2025], He et al. [2025] are forced to resort to computationally heavy test-time optimization Wang et al. [2023], Chen et al. [2025c], severely compromising real-time efficiency. While some attempts internalize physical priors during training to bypass this latency Zhong et al. [2025], their geometric penalties remain insufficient for dense clutters. In stark contrast, our framework elegantly resolves these bottlenecks: we capture the multimodal distribution via a Soft Winner-Takes-All objective, and directly bridge the precision gap using a human-inspired physical constraint during training, coupled with a physical ranking during inference.

3 Benchmark

3.1 Overview of Data Generation

Our data generation pipeline targets realistic cluttered scenes while producing two tightly coupled resources: (i) state observation representations; (ii) instruction-conditioned dexterous hand grasp pose labels. A central difficulty arises from the high DoF of dexterous hands: we must generate grasps that are not only physically feasible but also executable in cluttered scenes. Optimization-based methods have proven effective for dexterous hand grasp generation for a single object and can be

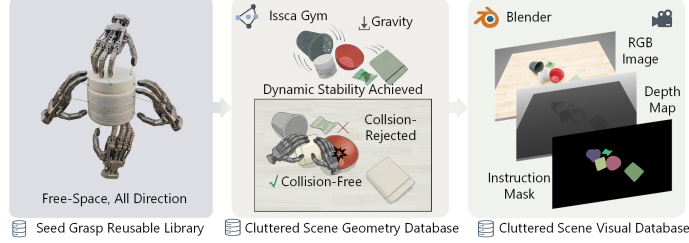


Figure 2: **Seed-and-filter generation pipeline.** Our two-stage framework generates scene-valid grasps by: (1) constructing a reusable seed grasp library in free space, and (2) simulating cluttered scenes in Isaac Gym for collision filtering, followed by multi-modal rendering in Blender.

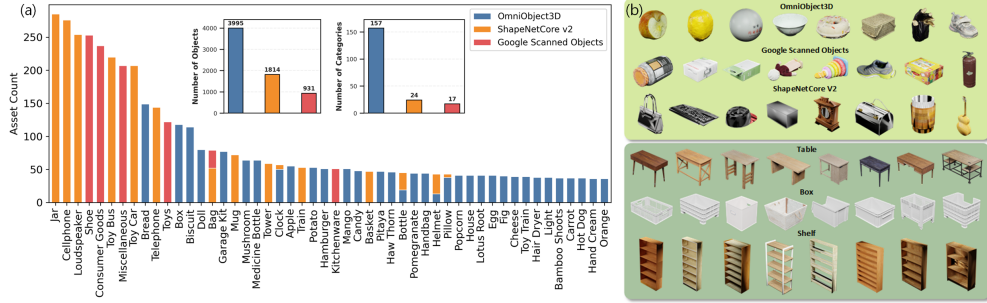


Figure 3: **Dataset statistics and visualization of 3D assets.** (a): Instance counts and category-wise distributions of the curated graspable objects, highlighting the top 50 semantic classes. (b): Visual examples of the assets, which range from diverse grasped objects (medicine bottles, boxes, apples) to realistic supporting bases (tables, open-top boxes, shelves).

119 applied in tabletop settings. However, the tabletop setting is inherently instance-specific, as each run
 120 is anchored to a particular object–table configuration. Moreover, its computational cost is prohibitive
 121 (e.g., optimizing 500 initializations for one object takes ~ 15 minutes on an A800), making per-object,
 122 per-scene optimization infeasible for large-scale cluttered scene benchmark construction.

123 As shown in Fig. 2, to enable scalable generation, we adopt a two-stage seed-and-filter strategy. First,
 124 we run optimization for each object in free space to produce a reusable library of seed grasp poses
 125 sampled from diverse directions. Next, we simulate cluttered scenes in Isaac Gym by dropping objects
 126 onto supporting bases. Once dynamic stability is achieved, we transform free-space seed grasp poses
 127 into scene-specific candidates and retain those that satisfy a conservative direct-retrieval criterion.
 128 Specifically, accepted candidates must establish target contact while avoiding support surfaces and
 129 clearly disruptive interactions with neighboring objects. This protocol is designed to isolate direct
 130 target extraction in cluttered scene, rather than rearrangement-mediated retrieval. To simulate the
 131 visual scene observations available to learning agents, we replay the cluttered scene geometry in
 132 Blender to build a visual database, including RGB images, depth maps, semantic segmentation, and
 133 camera parameters. The segmentation taxonomy is defined within each scene and used to generate
 134 target-object masks as visual instructions for grasping. In summary, the visual database provides
 135 diverse model inputs, while the geometry database provides scene-valid grasp ground truths for both
 136 model supervision and evaluation. Qualitative visualizations are in Appendix F.1.

137 3.2 3D Assets

138 **Grasped Objects.** We curate graspable object assets from OmniObject3D Wu et al. [2023], ShapeNet-
 139 Core v2 Chang et al. [2015], and Google Scanned Objects Downs et al. [2022] datasets. The com-
 140 position of the curated assets, including instance counts and category-wise distributions, is detailed
 141 in Fig. 3 (a). Our selection follows a staged protocol: we first use their semantic annotations to
 142 exclude categories incompatible with dexterous hand grasping, then enforce inter-class balancing to
 143 avoid long-tailed bias, and finally remove instances with defective meshes or incomplete textures.

Table 2: **Benchmark diversity and evaluation protocol across OmniDex subsets.** Each subset is organized with a large training split and two decoupled test tracks to evaluate generalization.

Subset	Obj. Div.	Scene Obj. Comp.	Sup. Base Type	Sup. Base Div.	Density	Scene Num.	Generalization Split		
							Train	Seen	Unseen
$2\mathcal{D}_{\text{std}}$	✓	Multi-category	Table	✗	0.1 / 0.2 / 0.3	1.14M	919K	62K	163K
$\mathcal{D}_{\text{mix}}$	✓	Multi-category	Table	✓	0.1 / 0.2 / 0.3	0.94M	706K	80K	152K
$\mathcal{D}_{\text{box}}$	✓	Multi-category	Box	✓	0.9	0.14M	112K	12K	16K
$\mathcal{D}_{\text{grid}}$	✓	Single-category	Table	✗	-	0.28M	246K	11K	24K
$\mathcal{D}_{\text{shelf}}$	✓	Multi-category	Shelf	✓	-	0.14M	107K	11K	28K

Each asset is processed into two scale-aligned representations: a physics-ready model generated via manifold-normalization-decomposition Wang et al. [2023] and a high-fidelity visual model, both sharing an identical normalization to ensure spatial consistency. In total, we provide a database of over 6,000 high-quality objects, spanning diverse categories such as cellphones, toys, and shoes. All selected instances are manually inspected to ensure they are suitable for the dexterous grasping task.

Supporting Base. As illustrated in Fig. 3 (b), our OmniDex benchmark incorporates tables, open-top boxes, and shelves to diversify object placement in cluttered scenes. Specifically, our asset library comprises 244 tables from ABO Collins et al. [2022], 79 open-top boxes sourced from both ABO and Isaac Lab Mittal et al. [2025], and 21 shelves from ShapeNetCore v2 Chang et al. [2015].

3.3 Scene-Specific Ground-Truth Grasp Pose Generation

Following the seed-and-filter strategy in Sec. 3.1, we first optimize 4,000 candidates per object using the force-closure estimator Wang et al. [2023], retaining those that pass a simulated lift test as seed grasp poses. To construct cluttered scenes, a random number of N objects are dropped from stochastic heights, spatially confined within the horizontal bounds of the supporting base. We adopt subset-specific scene generation rules to induce distinct clutter regimes: random tabletop clutter for $\mathcal{D}_{\text{std}}$ and $\mathcal{D}_{\text{mix}}$, dense confined stacking for $\mathcal{D}_{\text{box}}$, spaced identical-object layouts for $\mathcal{D}_{\text{grid}}$, and layer-wise arrangements for $\mathcal{D}_{\text{shelf}}$.

Simulation is terminated only after all objects satisfy stable resting conditions, with a fixed rollout cutoff to exclude non-convergent cases. The resulting settled scene yields stable object poses $\mathbf{o} = (\mathbf{t}_{\text{obj}}, \mathbf{R}_{\text{obj}})$, where $\mathbf{t}_{\text{obj}} \in \mathbb{R}^3$ and $\mathbf{R}_{\text{obj}} \in SO(3)$ denote the object translation and orientation, respectively. We then apply the object transformations to the selected seed grasps to derive scene-specific poses $\mathbf{g} = (\mathbf{t}_{\text{dex}}, \mathbf{R}_{\text{dex}}, \theta)$. Here, $\mathbf{t}_{\text{dex}} \in \mathbb{R}^3$ and $\mathbf{R}_{\text{dex}} \in SO(3)$ denote the wrist translation and orientation, respectively, while θ represents joint angles. The complete scene configuration is thus denoted as $\mathbf{s} = (\mathbf{o}, \mathbf{g})$. To mitigate artifacts from physical-visual mesh discrepancies and incomplete simulation convergence, we perform a sanity check by replaying each scene with visual models. Please see Appendix A.1 for more details.

3.4 Diversity and Protocol

Modal Diversity. As discussed in Sec. 1, geometry-only input is inadequate for instruction-guided grasping in cluttered scenes. We therefore provide aligned RGB, depth, instruction masks, and camera parameters under a subset-aware multi-view protocol. We use one ego-centric view and four auxiliary views by default, while adapting the camera layout to each subset: tabletop scenes use standard corner views, box scenes tighten the rig to counter boundary occlusion, and shelf scenes adopt layer-aligned frontal views to maintain visibility under severe vertical blockage. This makes perception a first-class component of the benchmark rather than a fixed rendering choice. Detailed camera settings are given in Appendix A.2.

Task Diversity. To align with the core objective of dexterous grasping, our OmniDex benchmark introduces diversity across: grasped object and supporting base. These factors directly dictate grasp feasibility and perceptual difficulty. As illustrated in Table 2, following prior works Ye et al. [2025], He et al. [2025], we provide $\mathcal{D}_{\text{std}}$ with 1.14M scenes where the supporting base remains fixed, varying only the target object. This design isolates the evaluation of policy transferability across diverse geometries and appearances. To evaluate grasping in more unstructured environments, we include $\mathcal{D}_{\text{mix}}$ with 0.94M scenes where both the target objects and the supporting bases vary simultaneously.

Real Scene Diversity. To approximate real-scene applications, we introduce three specialized $\mathcal{D}_{\text{box}}$, $\mathcal{D}_{\text{grid}}$, and $\mathcal{D}_{\text{shelf}}$. As illustrated in Fig. 1, $\mathcal{D}_{\text{box}}$ models parcel sorting, where objects are densely packed both vertically and horizontally within confined containers, mimicking carton or express package handling. $\mathcal{D}_{\text{grid}}$ simulates warehouse sorting environments featuring multiple identical objects uniformly distributed on a table. To ensure stable post-drop configurations in this subset, we restrict our asset pool to objects capable of remaining upright after falling (e.g., bottles). Finally, $\mathcal{D}_{\text{shelf}}$ targets SKU-level retail scenarios where products are systematically arranged on shelves; for this, we specifically curate retail-style items (e.g., noodle cups) from complete object dataset.

Benchmark Protocol. As summarized in Table 2, each subset is partitioned into a large training split and two test tracks: *Seen Objects in Unseen Layouts* and *Unseen Objects in Unseen Layouts*. The former isolates spatial reasoning under novel clutter configurations while controlling for object familiarity, whereas the latter probes zero-shot transfer to unseen geometries under equally novel layouts. This protocol cleanly separates layout novelty from object novelty, yielding a benchmark that measures robust scene-level generalization in cluttered dexterous grasping.

3.5 Metrics

Aligned with our OmniDex benchmark protocol, we evaluate model performance along two dimensions: physical grasp stability and protocol-consistent scene interaction under direct retrieval. Following Wang et al. [2023], a grasp is deemed physically valid if the target object securely withstands applied gravity. We quantify this via three Success Rate (SR) variants based on the tested gravitational directions: the actual scene gravity (SR_{scene} , our primary metric), all six canonical directions (SR_{any}), or at least one canonical direction (SR_{one}). To measure consistency with our selection protocol, we further report the Collision-Free Rate (CFR), defined as the percentage of generated grasps that avoid support-surface collisions and non-target contacts under the simulator.

4 Methodology

4.1 Overview

For clarity, we present the ego-centric single-view formulation here. We aim to generate a valid grasp pose $\hat{\mathbf{g}}$ for a target object in a cluttered scene, given an RGB image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, a depth map $\mathbf{D} \in \mathbb{R}^{H \times W \times 1}$, camera parameters $\{\mathbf{K}, \mathbf{T}\}$, the table height $\mathbf{H}$, and an instruction mask $\mathbf{M} \in \{0, 1\}^{H \times W}$. Here, the instruction mask explicitly specifies the target object within the clutter. $\mathbf{K}$ and $\mathbf{T}$ denote the intrinsic and extrinsic camera matrices, respectively, while H and W denote the image height and width. Our method is formulated as training a diffusion-based neural network to generate a high-DoF dexterous grasp pose conditioned on scene appearance, geometry, and target identity: $\hat{\mathbf{g}} = \Theta(\mathbf{I}, \mathbf{D}, \mathbf{M}, \mathbf{K}, \mathbf{T}, \mathbf{H})$. The multi-view extension is provided in Appendix C.2.

The overview of the OmniDex model is illustrated in Fig. 4. In the feature encoder module, we extract features from the camera image, depth map, and instruction mask, and aggregate them into a unified perception feature for the cluttered scene. It is subsequently combined with the scene parameter features to yield the condition embedding (Sec. 4.2). In the training phase, we apply a Soft Winner-Takes-All (WTA) loss to address the inherently multimodal distribution of grasp poses, alongside a novel physics-constrained loss designed to enforce close and stable picking (Sec. 4.3). In the inference phase, the model generates multiple grasp pose candidates, from which the optimal grasp is identified via a physical ranking and selection (Sec. 4.4).

4.2 Multimodal Scene-Aware Condition Embedding

To generate accurate and stable dexterous grasps in cluttered scenes, the model requires a comprehensive understanding of both the semantic and geometric structure of the workspace. Under the ego-centric single-view setting described above, we construct a multimodal, scene-aware condition embedding by fusing the RGB image, depth map, instruction mask, and scene parameters into a unified tokenized representation.

Given the ego-centric RGB image, we use a pre-trained DINOv2 Oquab et al. [2024] backbone to extract patch-level semantic features $\mathbf{F}_{\text{img}} \in \mathbb{R}^{N \times d}$, where N is the number of patches and d is the embedding dimension. We derive geometric features $\mathbf{F}_{\text{depth}} \in \mathbb{R}^{N \times d}$ from the depth map via a

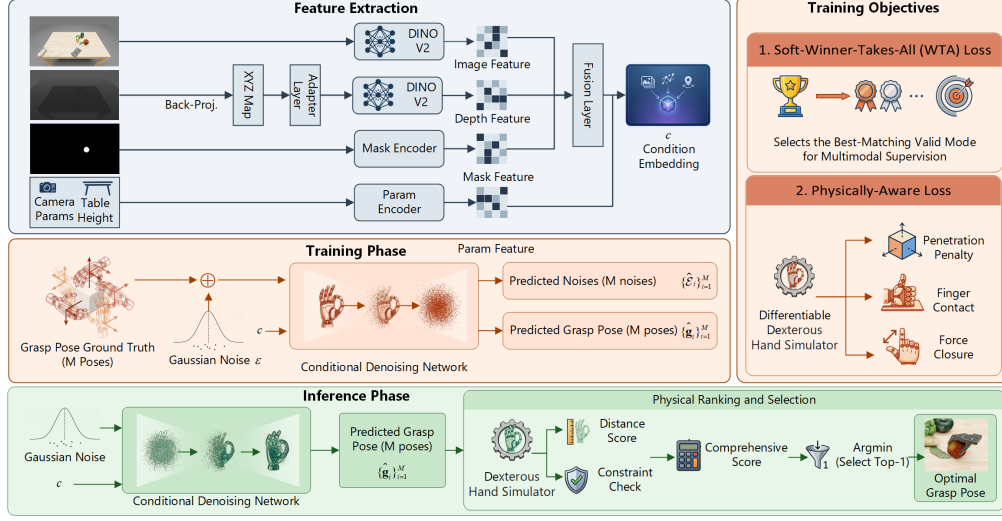


Figure 4: **Architecture of the OmniDex Model.** Condition embeddings are transformed from RGB-D inputs, instruction masks, and scene parameters. During training, a denoising network adds identical noise to M grasp poses, optimized jointly via a Soft Winner-Takes-All loss (for multimodal distributions) and a physics-aware loss (for stable contact). During inference, the network generates M candidates, followed by a physics-based ranking module to select the optimal grasp.

lightweight adapter, and encode the instruction mask into mask features $\mathbf{F}_{\text{mask}} \in \mathbb{R}^{N \times d}$. We then fuse semantic, geometric, and mask features to obtain a scene-aware representation $\mathbf{F}_{\text{scene}} \in \mathbb{R}^{N \times d}$. To anchor grasp generation in a consistent global frame, we additionally encode the camera parameters and table height into a parameter token $\mathbf{f}_{\text{para}} \in \mathbb{R}^{1 \times d}$. Appending $\mathbf{f}_{\text{para}}$ to $\mathbf{F}_{\text{scene}}$ yields the final condition embedding $\mathbf{c} \in \mathbb{R}^{(N+1) \times d}$. Implementation details are deferred to Appendix C.1.

4.3 Physically-Constrained Denoising for Multimodal Grasp Generation

Cluttered-scene grasping is intrinsically multimodal: under the same scene observation, a target object may admit multiple valid grasp modes. Deterministic supervision tends to average across these modes, often producing implausible intermediate poses. Inspired by winner-takes-all learning Rupprecht et al. [2017], we train the diffusion model with a soft WTA-style objective that preserves this one-to-many structure directly in grasping space. Specifically, we sample a set of M valid grasp poses for a target object and inject the same Gaussian noise ε into all M poses at diffusion step t . The network is then trained to explain the corrupted sample through the most compatible grasp mode rather than their average. Here, M serves as a finite hypothesis budget for approximating the dominant modes of the conditional grasp distribution. The resulting Soft-WTA loss is

$$\mathcal{L}_{\text{WTA}} = \mathbb{E}_{t, \varepsilon} \left[-\tau \log \sum_{i=1}^M \exp \left(-\|\varepsilon - \hat{\varepsilon}_i\|^2 / \tau \right) \right], \quad (1)$$

where $\hat{\varepsilon}_i$ is the predicted noise for the i -th grasp hypothesis and τ is a temperature parameter. This objective allows the model to represent diverse grasp modes without introducing handcrafted intermediate representations.

Purely data-driven denoising, however, is often insufficient for dexterous grasping, where millimeter-scale errors can break contact or induce unstable force closure. We therefore augment training with a physics-constrained objective defined on a differentiable dexterous hand simulator:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{WTA}} \mathcal{L}_{\text{WTA}} + \lambda_{\text{phys}} (\mathcal{L}_{\text{thumb}} + \mathcal{L}_{\text{opp}} + \mathcal{L}_{\text{SDF}} + \mathcal{L}_{\text{closure}}). \quad (2)$$

Here, $\mathcal{L}_{\text{thumb}}$ and $\mathcal{L}_{\text{opp}}$ encourage stable contact on the thumb and opposing fingers, $\mathcal{L}_{\text{closure}}$ promotes an antipodal grasp configuration, and $\mathcal{L}_{\text{SDF}}$ penalizes hand-object interpenetration. Full definitions are provided in Appendix C.3.

4.4 Physics-Driven Ranking and Selection

In the inference stage, a single sampled pose may exhibit geometric misalignment due to the inherent stochasticity of generative models. To enable robust execution, we introduce a parallel generation and physics-driven ranking paradigm. Specifically, given the same condition embedding $\mathbf{c}$, we simultaneously sample M independent Gaussian noises to decode candidate grasp poses $\{\hat{\mathbf{g}}_i\}_{i=1}^M$. These candidates are evaluated in the dexterous hand simulator to compute a comprehensive physical score $S(\hat{\mathbf{g}})$, assessing kinematic closure while strictly penalizing physical violations:

$$S(\hat{\mathbf{g}}) = S_{\text{kin}}(\hat{\mathbf{g}}) - R_{\text{con}}(\hat{\mathbf{g}}) + \Omega_{\text{phy}}(\hat{\mathbf{g}}) \quad (3)$$

The primary metric $S_{\text{kin}}(\hat{\mathbf{g}})$ evaluates force closure stability via finger-to-object distances as Sec. 4.3. The remaining terms act as essential constraint checks: the contact reward $R_{\text{con}}(\hat{\mathbf{g}})$ encourages multi-finger engagement by rewarding additional fingertip contacts, while the safety penalty Ω_{phy} linearly penalizes slight interpenetrations and heavily rejects severe collisions or missing thumb-opposition. Ultimately, we filter out physically invalid modes to output the optimal grasp pose.

5 Experiment

5.1 Implementation Details

We instantiate our framework on the Shadow Dexterous Hand Shadow Robot Company [2005], leveraging its high-dimensional action space (24 DoF) to rigorously validate our model’s generative capacity. All subsets of our proposed benchmark adopt an 8/2 split for the train and test sets. The test sets are decoupled into two distinct tracks: Seen objects in unseen clustered scene; unseen objects (see Appendix B). Please refer to Appendix D for more details.

5.2 Main Results

As shown in Table 3, our method establishes the state-of-the-art in cluttered-scene dexterous grasping. On $\mathcal{D}_{\text{std}}$ with ego-view input, OmniDex achieves 55.80 SR_{scene} and 84.12 CFR, surpassing Dex-GraspVLA by 14.81 and 2.68 points, respectively. This advantage also holds for unseen objects, where OmniDex attains 59.74 SR_{scene} , showing strong generalization beyond the training set. Across subsets, our method remains effective under varying scene layouts and support types, and reaches particularly strong performance on $\mathcal{D}_{\text{grid}}$, suggesting that our model can reliably exploit structured spatial cues. Overall, these results verify that the OmniDex model delivers superior accuracy, robustness, and cross-setting generalization for realistic cluttered-scene grasping. Qualitative results are shown in Appendix F.2.

Table 3: Main Results on the OmniDex Benchmark. We evaluate our method against baselines across diverse scene subsets and compare with other methods on $\mathcal{D}_{\text{std}}$. * denotes the use of the finger-force trick Wang et al. [2023]

Method	Subset	Input View	Seen Objects							Unseen Objects						
			SR_{scene}	$\text{SR}_{\text{scene}}^*$	SR_{any}	SR_{any}^*	SR_{one}	SR_{one}^*	CFR	SR_{scene}	$\text{SR}_{\text{scene}}^*$	SR_{any}	SR_{any}^*	SR_{one}	SR_{one}^*	CFR
Ours	$\mathcal{D}_{\text{std}}$	Ego	33.60	55.80	18.87	36.79	65.15	79.35	84.12	40.15	59.74	27.18	40.77	65.54	81.77	84.61
	$\mathcal{D}_{\text{mix}}$	Ego	32.05	52.21	17.12	34.04	58.51	74.52	73.12	37.90	55.10	24.13	35.19	62.41	78.02	83.22
	$\mathcal{D}_{\text{box}}$	Ego	32.34	35.83	17.06	19.79	55.15	56.52	99.12	33.48	36.50	17.80	20.17	56.77	58.11	95.72
	$\mathcal{D}_{\text{grid}}$	Ego	55.01	79.60	33.68	55.02	91.91	93.70	95.46	43.11	68.05	33.97	52.67	87.50	92.09	94.86
	$\mathcal{D}_{\text{shelf}}$	Ego	29.49	39.46	18.31	25.25	55.17	60.57	81.95	14.96	24.08	11.58	15.08	34.26	37.89	63.57
	$\mathcal{D}_{\text{std}}$	Multi	37.35	60.65	22.18	41.18	68.01	82.60	85.11	45.05	64.08	30.79	43.76	68.48	84.50	87.24
	$\mathcal{D}_{\text{mix}}$	Multi	35.41	56.02	20.54	38.31	62.47	78.06	75.31	42.56	60.34	28.36	40.12	65.83	82.61	85.04
	$\mathcal{D}_{\text{box}}$	Multi	27.89	28.52	19.11	19.53	41.20	42.06	97.91	31.14	31.71	21.17	21.79	43.33	44.30	97.64
	$\mathcal{D}_{\text{grid}}$	Multi	60.37	84.28	38.44	58.95	95.14	96.52	96.49	46.68	73.14	37.42	56.82	91.21	95.31	95.71
	$\mathcal{D}_{\text{shelf}}$	Multi	34.06	44.63	22.27	29.61	59.88	64.31	83.42	18.21	28.86	15.24	19.37	39.54	42.21	65.29
DexGraspVLA Zhong et al. [2026]	$\mathcal{D}_{\text{std}}$	Ego	24.30	40.99	11.87	24.42	53.83	67.05	81.44	39.42	54.97	29.95	41.02	54.89	71.55	42.86
DexGraspNet 2.0 Zhang et al. [2024]	$\mathcal{D}_{\text{std}}$	Ego	28.03	43.64	13.16	25.59	56.04	68.47	81.06	32.40	47.60	18.83	27.39	58.77	73.73	85.15
Grasp-as-You-Say Wei et al. [2024]	$\mathcal{D}_{\text{std}}$	Ego	0.24	1.68	0.09	0.15	20.91	27.45	38.57	0.44	2.55	0.10	0.42	19.46	25.57	64.25

5.3 Ablation Studies

We evaluate our procedures on a representative mini-split curated from $\mathcal{D}_{\text{std}}$ (see Appendix B). By default, baselines operate on a single ego-centric view. We use SR_{scene} as the principal metric to quantify grasp stability. More ablation studies can be seen in Appendix E.

Effect of Benchmark Scale. To justify the massive scale of our proposed OmniDex benchmark, we train our model on incrementally larger fractions of the training set, and evaluate it on the full test set. As Table 4 shows, performance scales strictly with data volume. Scaling explicitly enhances the model’s capacity to reason about complex inter-object physical interactions in cluttered layouts; notably, CFR surges from 81.54% (20% subset) to 83.66% (100% subset). This significant improvement in stable grasping validates the necessity of our large-scale data generation pipeline.

Table 4: Quantitative results when trained on incrementally larger fractions of training dataset.

Metrics	20%	40%	60%	80%	100%
SR _{scene}	40.50	41.04	43.84	44.32	44.41
CFR	81.54	82.64	82.13	84.15	83.66

Effect of Input Modalities. We analyze the role of various inputs as shown in Table 5. Depth provides the foundational shape cognition. RGB supplies scene semantics and, more crucially, localizes the target object within the cluster. Camera parameters ensure accurate geometric perception. Multi-view inputs yield significant performance gains by enabling comprehensive scene understanding. Table height serves as a vital spatial anchor; since the coordinate origin lies at the table center, it aligns the depth-derived geometry with grasp ground truths to explicitly dictate precise grasp locations.

Table 5: Quantitative ablation on various input conditions. The highlighted row represents baseline.

RGB	Depth	Cam. Para.	Table Height	Multi-view	SR _{scene}	CFR
✓		✓	✓		31.89	79.97
	✓	✓	✓		36.80	82.81
✓	✓		✓		43.68	84.09
✓	✓	✓			40.64	84.04
✓	✓	✓	✓	✓	49.87	82.11
✓	✓	✓	✓		44.41	83.66

Effect of Model Components. Table 6 details our architectural ablations. Standard denoising loss traps generation in an averaging effect across grasp hypotheses. Conversely, $\mathcal{L}_{\text{WTA}}$ resolves this multi-modal manifold issue, substantially boosting SR_{scene}. Training-phase physical constraints are vital for overcoming the critical last-millimeter contact challenge. At inference, our physics-driven ranking drastically outperforms naive random sampling, elevating SR_{scene} from 30% to 40%. Test-time optimization yields limited gains, bottlenecked by inherent instability and depth-backprojection occlusions (see details in Appendix E).

Table 6: Component-wise ablation study evaluated on both training and inference phases. TTP denotes Test-Time Optimization.

Training Phase						Inference Phase			Metrics	
$\mathcal{L}_{\text{stand}}$	$\mathcal{L}_{\text{WTA}}$	$\mathcal{L}_{\text{SDF}}$	$\mathcal{L}_{\text{thumb}}$	$\mathcal{L}_{\text{opp}}$	$\mathcal{L}_{\text{closure}}$	Naive Samp.	Rank.	TTP	SR _{scene}	CFR
✓							✓		37.79	81.37
	✓						✓		38.07	81.31
	✓	✓					✓		41.93	84.06
	✓	✓	✓				✓		43.19	84.40
	✓	✓	✓	✓		✓			36.08	79.71
	✓	✓	✓	✓	✓			✓	36.16	2.56
	✓	✓	✓	✓	✓		✓		44.41	83.66

6 Conclusion

In this paper, we presented OmniDex to tackle the data and algorithmic challenges of dexterous grasping in cluttered scenes. Our OmniDex benchmark leverages a scalable seed-and-filter strategy to deliver over 2.6 million scenes and 0.4 billion validated grasp ground truths, complete with rich multi-modal observations. Built upon this, our novel generation framework couples Soft Winner-Takes-All learning with human-inspired physical constraints to effectively capture grasp multimodality and bridge the crucial last-millimeter precision gap. Experimental results confirm that our OmniDex model achieves state-of-the-art performance on the proposed benchmark, significantly outperforming existing baselines on standard cluttered scenes.

References

- Jiafei Duan, Samson Yu, Hui Li Tan, Hongyuan Zhu, and Cheston Tan. A survey of embodied ai: From simulators to research tasks. *IEEE Trans. Emerg. Top. Comput. Intell.*, 6(2):230–244, 2022.
- Hao-Shu Fang, Chenxi Wang, Minghao Gou, and Cewu Lu. Graspnet-1billion: A large-scale benchmark for general object grasping. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 11444–11453, 2020.
- Hao-Shu Fang, Chenxi Wang, Hongjie Fang, Minghao Gou, Jirong Liu, Hengxu Yan, Wenhai Liu, Yichen Xie, and Cewu Lu. Anygrasp: Robust and efficient grasp perception in spatial and temporal domains. *IEEE Trans. Robot.*, 39(5):3929–3945, 2023.
- Yumeng Liu, Yaxun Yang, Youzhuo Wang, Xiaofei Wu, Jiamin Wang, Yichen Yao, Sören Schwertfeger, Sibe Yang, Wenping Wang, Jingyi Yu, et al. Realdex: towards human-like grasping for robotic dexterous hand. In *Proc. Int. Joint Conf. Artif. Intell.*, pages 6859–6867, 2024.
- Jianglong Ye, Keyi Wang, Chengjing Yuan, Ruihan Yang, Yiquan Li, Jiyue Zhu, Yuzhe Qin, Xueyan Zou, and Xiaolong Wang. Dex1b: Learning with 1b demonstrations for dexterous manipulation. In *RSS Workshop on Dexterous Manipulation*, 2025.
- Jiayi Chen, Yubin Ke, Lin Peng, and He Wang. Dexonomy: Synthesizing all dexterous grasp types in a grasp taxonomy. In *RSS Workshop on Dexterous Manipulation*, 2025a.
- Ruicheng Wang, Jialiang Zhang, Jiayi Chen, Yinzhen Xu, Puhao Li, Tengyu Liu, and He Wang. Dexgraspnet: A large-scale robotic dexterous grasp dataset for general objects based on simulation. In *IEEE Int. Conf. Robot. Autom.*, pages 11359–11366. IEEE, 2023.
- Yinzhen Xu, Weikang Wan, Jialiang Zhang, Haoran Liu, Zikang Shan, Hao Shen, Ruicheng Wang, Haoran Geng, Yijia Weng, Jiayi Chen, et al. Unidexgrasp: Universal robotic dexterous grasping via learning diverse proposal generation and goal-conditioned policy. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 4737–4746, 2023.
- Jialiang Zhang, Haoran Liu, Danshi Li, XinQiang Yu, Haoran Geng, Yufei Ding, Jiayi Chen, and He Wang. Dexgraspnet 2.0: Learning generative dexterous grasping in large-scale synthetic cluttered scenes. In *Conf. Robot. Learn.*, 2024.
- Jiyao Zhang, Zhiyuan Ma, Tianhao Wu, Zeyuan Chen, and Hao Dong. Cadgrasp: Learning contact and collision aware general dexterous grasping in cluttered scenes. In *Adv. Neural Inf. Process. Syst.*, 2025.
- Yi-Lin Wei, Mu Lin, Yuhao Lin, Jian-Jian Jiang, Xiao-Ming Wu, Ling-An Zeng, and Wei-Shi Zheng. Afford-dexgrasp: Open-set language-guided dexterous grasp with generalizable-instructive affordance. In *Int. Conf. Comput. Vis.*, pages 11818–11828, 2025.
- Jens Lundell, Francesco Verdoja, and Ville Kyrki. Ddgc: Generative deep dexterous grasping in clutter. *IEEE Robot. Autom. Lett.*, 6(4):6899–6906, 2021.
- Zeyuan Chen, Qiyang Yan, Yuanpei Chen, Tianhao Wu, Jiyao Zhang, Zihan Ding, Jinzhou Li, Yaodong Yang, and Hao Dong. Clutterdexgrasp: A sim-to-real system for general dexterous grasping in cluttered scenes. In *Conf. Robot. Learn.*, pages 885–905. PMLR, 2025b.
- Jiawei He, Danshi Li, Xinqiang Yu, Zekun Qi, Wenyao Zhang, Jiayi Chen, Zhaoxiang Zhang, Zhizheng Zhang, Li Yi, and He Wang. Dexvlg: Dexterous vision-language-grasp model at scale. In *Int. Conf. Comput. Vis.*, pages 14248–14258, 2025.
- Youzhuo Wang, Jiayi Ye, Chuyang Xiao, Yiming Zhong, Heng Tao, Hang Yu, Yumeng Liu, Jingyi Yu, and Yuexin Ma. Dexh2r: A benchmark for dynamic dexterous grasping in human-to-robot handover. In *Int. Conf. Comput. Vis.*, pages 12702–12712, 2025.
- Mengda Xu, Han Zhang, Yifan Hou, Zhenjia Xu, Linxi Fan, Manuela Veloso, and Shuran Song. Dexumi: Using human hand as the universal manipulation interface for dexterous manipulation. In *Workshop on Human to Robot*, 2025.
- Jinglue Hang, Xiangbo Lin, Tianqiang Zhu, Xuanheng Li, Rina Wu, Xiaohong Ma, and Yi Sun. Dexfuncgrasp: A robotic dexterous functional grasp dataset constructed from a cost-effective real-simulation annotation system. In *AAAI*, volume 38, pages 10306–10313, 2024.
- Yiming Zhong, Qi Jiang, Jingyi Yu, and Yuexin Ma. Dexgrasp anything: Towards universal robotic dexterous grasping with physics awareness. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 22584–22594, 2025.

370 Weikang Wan, Haoran Geng, Yun Liu, Zikang Shan, Yaodong Yang, Li Yi, and He Wang. Unidexgrasp++:
371 Improving dexterous grasping policy learning via geometry-aware curriculum and iterative generalist-specialist
372 learning. In *Int. Conf. Comput. Vis.*, pages 3891–3902, 2023.

373 Yuzhe Qin, Binghao Huang, Zhao-Heng Yin, Hao Su, and Xiaolong Wang. Dexpoint: Generalizable point
374 cloud reinforcement learning for sim-to-real dexterous manipulation. In *Conf. Robot. Learn.*, pages 594–605.
375 PMLR, 2023.

376 Jiayi Chen, Yubin Ke, and He Wang. Bodex: Scalable and efficient robotic dexterous grasp synthesis using
377 bilevel optimization. In *IEEE Int. Conf. Robot. Autom.*, pages 01–08. IEEE, 2025c.

378 Tong Wu, Jiarui Zhang, Xiao Fu, Yuxin Wang, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen
379 Qian, et al. Omniobject3d: Large-vocabulary 3d object dataset for realistic perception, reconstruction and
380 generation. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 803–814, 2023.

381 Angel X Chang, Thomas Funkhouser, Leonidas Guibas, Pat Hanrahan, Qixing Huang, Zimo Li, Silvio Savarese,
382 Manolis Savva, Shuran Song, Hao Su, et al. Shapenet: An information-rich 3d model repository. *arXiv*
383 *preprint arXiv:1512.03012*, 2015.

384 Laura Downs, Anthony Francis, Nate Koenig, Brandon Kinman, Ryan Hickman, Krista Reymann, Thomas B
385 McHugh, and Vincent Vanhoucke. Google scanned objects: A high-quality dataset of 3d scanned household
386 items. In *IEEE Int. Conf. Robot. Autom.*, pages 2553–2560, 2022.

387 Jasmine Collins, Shubham Goel, Kenan Deng, Achleshwar Luthra, Leon Xu, Erhan Gundogdu, Xi Zhang, Tomas
388 F Yago Vicente, Thomas Dideriksen, Himanshu Arora, et al. Abo: Dataset and benchmarks for real-world 3d
389 object understanding. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 21126–21136, 2022.

390 Mayank Mittal, Pascal Roth, James Tigue, Antoine Richard, Octi Zhang, Peter Du, Antonio Serrano-Munoz,
391 Xinjie Yao, René Zurbügg, Nikita Rudin, et al. Isaac lab: A gpu-accelerated simulation framework for
392 multi-modal robot learning. *arXiv preprint arXiv:2511.04831*, 2025.

393 Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez,
394 Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without
395 supervision. *Trans. Mach. Learn. Res.*, 2024.

396 Christian Rupprecht, Iro Laina, Robert DiPietro, Maximilian Baust, Federico Tombari, Nassir Navab, and
397 Gregory D Hager. Learning in an uncertain world: Representing ambiguity through multiple hypotheses. In
398 *Int. Conf. Comput. Vis.*, pages 3591–3600, 2017.

399 Shadow Robot Company. Shadow dexterous hand. <https://www.shadowrobot.com/dexterous-hand/>,
400 2005.

401 Yifan Zhong, Xuchuan Huang, Ruochong Li, Ceyao Zhang, Zhang Chen, Tianrui Guan, Fanlian Zeng, Ka Nam
402 Lui, Yuyao Ye, Yitao Liang, et al. Dexgraspvla: A vision-language-action framework towards general
403 dexterous grasping. In *AAAI*, volume 40, pages 18836–18844, 2026.

404 Yi-Lin Wei, Jian-Jian Jiang, Chengyi Xing, Xian-Tuo Tan, Xiao-Ming Wu, Hao Li, Mark Cutkosky, and Wei-Shi
405 Zheng. Grasp as you say: Language-guided dexterous grasp generation. volume 37, pages 46881–46907,
406 2024.

407 Leonid Keselman, John Iselin Woodfill, Anders Grunnet-Jepsen, and Achintya Bhowmik. Intel realsense
408 stereoscopic depth cameras. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 1–10, 2017.

409 Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*,
410 2017.

A Benchmark Generation Pipeline Details

A.1 Cluttered Scene Geometry Database Generation

Supporting Base Asset Setting. All supporting structures are canonicalized to a gravity-aligned orientation. We standardize their local coordinate frames by placing the origin at the intersection of the gravity axis and the ground plane, facilitating straightforward base anchoring within the simulation environment.

Simulation Setup and Physical Properties. We leverage Isaac Gym4 as our core physics engine to simulate clustered scene. To ensure stable and uniform dynamic behaviors across varied shapes, all object instances are assigned universally consistent physical properties: a density of 500 kg/m³, coupled with linear and angular damping factors of 1.0 and 0.3, respectively.

Subset-Specific Initialization Strategies. To systematically decouple the generalization evaluation, we employ tailored spatial initialization logic for each subset:

- $\mathcal{D}_{\text{std}}$ and $\mathcal{D}_{\text{mix}}$: We randomly sample 2 to 7 unique object instances. Objects are dropped from a height of 0.5m above the table and their dropping horizontal spawn coordinates are tightly bounded within the table’s footprint.

- $\mathcal{D}_{\text{box}}$: 2 to 10 unique instances are initialized 0.4m above the top rim of the box. The horizontal dropping area is deliberately constricted to a 0.8 scale of the box’s extent, naturally inducing dense and complex clustered stacking.

- $\mathcal{D}_{\text{grid}}$: We replicate 4 to 10 identical instances. To simulate structured arrays, objects are uniformly distributed within a 0.8 scaled tabletop area. Layouts are dynamically computed utilizing object bounding boxes, enforcing a strict minimum inter-instance spacing of 5cm.

- $\mathcal{D}_{\text{shelf}}$: Based on explicitly annotated layer heights, objects are uniformly arranged along each shelf layer using their bounding boxes, maintaining a minimum spacing of 4cm.

Dynamic Resting Conditions. The simulation evolves until the scene reaches a physically stable resting state. We define termination by two rigorous criteria: (1) kinematic energy dissipation, where the linear and angular velocities of all objects fall below 0.03 m/s and 0.10 rad/s, respectively; or (2) reaching a hard cutoff at 500 simulation steps to prevent infinite oscillation.

Rigorous Geometric Validation. To completely eliminate physically anomalous artifacts commonly found in automated generation pipelines, we implement a strict post-simulation geometric sanity check. A generated scene is systematically rejected unless it satisfies the following tolerance thresholds:

- Sinking & Floating:** The maximum allowed vertical deviation (embedding into the base or hovering above it) is strictly bounded to 3mm.

- Inter-object Penetration:** The maximum permissible mutual penetration depth between any two clustered objects is capped at 2mm.

Tolerance in Contact Filtering. Our filtering is conservative but not perfectly zero-tolerance at the contact level. In practice, small numerical overlaps may arise from mesh decomposition, contact discretization, and simulator tolerances. We therefore allow a limited penetration margin during filtering and reject only clearly invalid grasp poses. In the current implementation, object–support penetration is tolerated up to 2cm before rejection, while shallower overlaps are retained to reduce false negatives caused by imperfect collision geometry.

A.2 Cluttered Scene Visual Database Generation

We design a multi-camera rig setup that mimics typical real-world robotic workspaces. We deploy five fixed Intel RealSense D455 sensors Keselman et al. [2017]. To maximize the visual coverage of complex object interactions and mitigate severe occlusions, the camera poses are adaptively configured based on the subset:

- $\mathcal{D}_{\text{std}}$, $\mathcal{D}_{\text{mix}}$, $\mathcal{D}_{\text{grid}}$: For scenes without pronounced boundary occlusions, the primary ego-centric camera is positioned 0.6m above the global coordinate origin and retracted 0.3m outward from the table edge. Auxiliary multi-view cameras are symmetrically deployed 0.25m outward from the table

460 corners. To guarantee optimal object framing, the optical axes of all cameras are explicitly oriented
461 towards a focal point 0.15m below the spatial center.

462 • $\mathcal{D}_{\text{box}}$: The high physical boundaries of the container in the box subset introduce severe peripheral
463 occlusions. To counteract this, we dynamically tighten the camera rig. The ego-centric camera and
464 the auxiliary corner cameras are moved closer, fixed at a uniform 0.2m outward offset from the table
465 edges and corners.

466 • $\mathcal{D}_{\text{shelf}}$: The multi-layered geometry of shelf imposes severe vertical visibility constraints, rendering
467 standard top-down or steep perspectives entirely ineffective due to overhanging structures. We deploy
468 a specialized 5-camera rig driven by a dynamic, layer-wise elevation-tracking strategy. The z -axis
469 coordinates of all five cameras and their focal targets are aligned with the height of shelf, establishing
470 a zero-pitch, horizon-aligned viewing frustum. The primary ego-centric camera is positioned squarely
471 in front of the shelf, with its standoff distance adaptively scaled by a factor of 1.2 of the shelf’s
472 spatial diagonal. Simultaneously, the four auxiliary cameras are deployed radially outward from the
473 top-surface corners, pushed along the diagonal axes by a scaling factor of 0.8.

474 B Benchmark Split and Density Details

475 To comprehensively assess grasping stability under varying degrees of physical entanglement, we
476 deliberately modulate the cluster density, particularly for the $\mathcal{D}_{\text{std}}$ and $\mathcal{D}_{\text{mix}}$ subsets. This is explicitly
477 achieved by constraining the object dropping area during simulation as a scalar ratio of the full
478 tabletop extent. Specifically, we synthesize scenes across three distinct density tiers (ratios of 0.1,
479 0.2, and 0.3), with each density level contributing approximately 0.6M scenes to the overarching
480 benchmark.

481 For the ablation mini-split, we uniformly sample $\sim 10\%$ of the training and testing scenes across
482 all density tiers, yielding a diverse subset of roughly 90K scenes. This highly challenging spatial
483 configuration allows us to rigorously benchmark the geometric reasoning capabilities of individual
484 model components under dense clutter.

485 C OmniDex Model Details

486 C.1 Single-View Condition Embedding

487 For the ego-centric single-view setting, we extract patch-level semantic features from the RGB image
488 using a pre-trained DINOv2 Oquab et al. [2024] backbone, yielding $\mathbf{F}_{\text{img}} \in \mathbb{R}^{N \times d}$, where N is the
489 number of patches and d is the embedding dimension. To encode scene geometry, we back-project
490 the depth map into a 3D coordinate map $\mathbf{X} \in \mathbb{R}^{H \times W \times 3}$:

$$\mathbf{X}(u, v) = \mathbf{D}(u, v) \mathbf{K}^{-1} [u, v, 1]^T, \quad (4)$$

491 where (u, v) denotes the 2D pixel coordinate. We then use a lightweight adapter to map the XYZ
492 representation into the DINOv2 input space, producing geometric features $\mathbf{F}_{\text{depth}} \in \mathbb{R}^{N \times d}$.

493 To isolate the target object from clutter, the instruction mask is encoded into mask features
494 $\mathbf{F}_{\text{mask}} \in \mathbb{R}^{N \times d}$. We concatenate semantic, geometric, and mask features along the channel di-
495 mension and fuse them into a scene-aware representation $\mathbf{F}_{\text{scene}} \in \mathbb{R}^{N \times d}$.

496 To preserve a consistent global reference for grasp generation, we additionally encode the camera
497 parameters and table height. Specifically, we compute the projection matrix $\mathbf{P} = \mathbf{K}\mathbf{T}$ and encode
498 $\mathbf{P}$ together with $\mathbf{H}$ via MLPs to obtain a parameter token $\mathbf{f}_{\text{para}} \in \mathbb{R}^{1 \times d}$. Appending $\mathbf{f}_{\text{para}}$ to $\mathbf{F}_{\text{scene}}$
499 yields the final condition embedding $\mathbf{c} \in \mathbb{R}^{(N+1) \times d}$.

500 C.2 Multi-View Input Fusion

501 To effectively process extreme spatial occlusions, we introduce a hybrid and physics-aware encoder for
502 multi-view observation. Instead of relying on a monolithic black-box transformer to indiscriminately
503 mix multi-view input, our architecture deliberately decouples the fusion paradigm into two distinct
504 pathways: a data-driven semantic early RGB fusion, a rigorous geometry-aware depth reprojection,
505 and camera relationship tokens.

Data-Driven Early RGB Fusion. Traditional multi-view encoders often suffer from computational overhead when processing independent visual tokens across views. To ensure efficiency without losing critical textural semantics, we concatenate the multi-view RGB, depth, and instance masks, feeding them into a lightweight convolutional adapter. This yields an inter-view alpha map via a softmax operation across the view dimension V . The RGB inputs are then aggregated using this learned alpha map, effectively allowing the network to dynamically stitch unoccluded visual patches from auxiliary cameras into a cohesive representation.

Rigorous Geometry-Aware Depth Reprojection. Unlike textural features, depth represents strict spatial geometry that cannot be naively interpolated. To capture the 3D manifold of the cluttered scene, we explicitly route the multi-view depth maps to the primary anchor camera using strict camera parameters. To maintain computational tractability, this point-cloud inversion is performed utilizing a hardware-accelerated Z-buffer scattering mechanism. The physically fused depth map is subsequently converted into a normalized XYZ spatial map.

Camera Relationship Tokens. To supplement the dense visual features, our encoder explicitly injects global contextual priors. The camera poses are tokenized and processed through a View Relation Transformer to capture the spatial topology of the multi-camera rig. These auxiliary tokens are concatenated with the visual features to form the final comprehensive observation sequence for the downstream generative policy.

524 C.3 Physically-Constrained Loss

We first back-project the depth map and filter it utilizing the instruction mask to extract a partial 3D point cloud of the targeted object, denoted as $\mathcal{P}_{\text{obj}}$. Simultaneously, we obtain the predicted grasp pose $\hat{\mathbf{g}}$ from the denoised output and feed it into the differentiable dexterous hand simulator to compute the predefined contact points on the hand surface Wang et al. [2023], denoted as $\mathcal{P}_{\text{con}}$. For any point $p \in \mathcal{P}_{\text{con}}$, its minimal distance to the object point cloud is defined as $d(p) = \min_{q \in \mathcal{P}_{\text{obj}}} \|p - q\|_2$.

Effective grasping strictly requires a stable kinematic clamping force. Morphologically, the thumb plays a dominant role in opposing the other fingers. We explicitly enforce this by partitioning the contact points into a thumb set $\mathcal{P}_{\text{thumb}}$ and an opposing fingers set $\mathcal{P}_{\text{opp}}$. Specifically, we apply a thumb contact loss as:

$$\mathcal{L}_{\text{thumb}} = \max \left(\frac{1}{|\mathcal{P}_{\text{thumb}}|} \sum_{p \in \mathcal{P}_{\text{thumb}}} d(p) - \epsilon_{\text{contact}}, 0 \right) \quad (5)$$

534 where $\epsilon_{\text{contact}}$ is a contact tolerance margin.

535 For the opposing fingers, forming a stable force closure often only requires a partial subset to make
536 contact. Therefore, we apply an identical loss formulation for $\mathcal{L}_{\text{opp}}$, but compute the average distance
537 over exclusively the K closest points within $\mathcal{P}_{\text{opp}}$.

538 Beyond simple proximity, a stable grasp necessitates a valid force closure, requiring the fingers to
539 approach the object from opposing directions. To explicitly encourage this antipodal grasping pattern,
540 we introduce a force closure loss as:

$$\mathcal{L}_{\text{closure}} = \max (\cos (v_{\text{thumb}}, v_{\text{opp}}) + \epsilon_{\text{closure}}, 0) \quad (6)$$

541 where $v_{\text{thumb}}, v_{\text{opp}}$ are approach vectors from finger to object center, $\cos(\cdot, \cdot)$ denotes the cosine
542 similarity, and $\epsilon_{\text{closure}}$ is a tolerance margin.

543 Finally, to prevent physically impossible interpenetrations between the hand and the target object,
544 we incorporate a Signed Distance Field (SDF) penetration loss $\mathcal{L}_{\text{SDF}}$. Let $\Phi_{\text{obj}}(\cdot)$ denote the SDF of
545 the object, where negative values indicate spatial locations strictly inside the object’s geometry. We
546 penalize any hand mesh vertices $\mathcal{P}_{\text{hand}}$ that penetrate the object as follows:

$$\mathcal{L}_{\text{SDF}} = \frac{1}{|\mathcal{P}_{\text{hand}}|} \sum_{p \in \mathcal{P}_{\text{hand}}} \max(-\Phi_{\text{obj}}(p), 0) \quad (7)$$

where $\mathcal{P}_{\text{hand}}$ represents the set of all vertices on the predicted hand mesh. This term effectively repels the hand fingers and palm from intersecting with the object’s interior volume.

D Supplementary Implementation Details

All experiments are conducted on NVIDIA A800 GPUs. The generation of the cluster scene geometry database and the cluster scene visual database consumed about 5K and 17K GPU hours, respectively. The AdamW optimizer Loshchilov and Hutter [2017] is used with an initial learning rate of e^{-4} and a weight decay of e^{-4} . All training durations are set to 25 epochs. For both the STA loss optimization during training and the physics-driven ranking strategy during inference, we uniformly set the number of parallel grasp hypotheses to $M = 64$. We define loss weights $\lambda_{\text{WTA}} = 1$ and $\lambda_{\text{phys}} = 100$, respectively. Within the physical constraint formulation, the relative weighting among the sub-components is specifically calibrated as $\mathcal{L}_{\text{thumb}} : \mathcal{L}_{\text{opp}} : \mathcal{L}_{\text{SDF}} : \mathcal{L}_{\text{closure}} = 5 : 1 : 0.2 : 2$. Notably, we assign a substantially higher penalty weight to the thumb, explicitly reflecting its indispensable kinematic role in establishing robust force closure. Imposing strict physical penalties on an uninitialized network often leads to severe learning instability. To mitigate this, we employ a delayed linear warm-up scheduling for the physical constraints. Specifically, the physical loss term is entirely deactivated ($\lambda_{\text{phys}} = 0$) during the initial 10 epochs, allowing the generative network to first learn a stable grasp manifold. Subsequently, λ_{phys} is linearly ramped up to its maximum target value of 100 between epochs 10 and 20, ensuring a smooth transition towards physically compliant grasp synthesis.

Our benchmark is constructed from third-party assets obtained from their official releases, including OmniObject3D Wu et al. [2023], ShapeNetCore v2 Chang et al. [2015], Google Scanned Objects Downs et al. [2022], ABO Collins et al. [2022], and Isaac Lab Mittal et al. [2025]. We credit these original resources in the paper and use them in accordance with their respective licenses and terms of use. In particular, OmniObject3D and Google Scanned Objects are released under CC BY 4.0, ABO is released under CC BY 4.0, ShapeNetCore v2 is used under the ShapeNet Terms of Use for non-commercial research, and Isaac Lab is distributed under the BSD-3-Clause license.

E Supplementary Ablation Study

To evaluate the efficiency-performance trade-off during inference, we ablate M , the number of parallel generated grasp hypotheses, and comparison with TTP. For a rigorously fair comparison, the TTP baseline is configured to optimize the exact same physical loss objectives utilized in the training phase, executing for 200 steps per grasp. Quantitative results are detailed in Table D1.

Table D1: Effect of parallel hypothesis generation vs. Test-Time Optimization.

Metric	Parallel Hypotheses (M)							TTP
	1	2	4	8	16	32	64	
SR _{scene}	36.08	38.02	41.06	42.47	43.41	44.37	44.41	36.16
CFR	79.71	80.81	81.58	82.26	82.62	83.09	83.66	2.56
Inference Time (ms)	219.14	233.35	266.39	325.20	543.93	948.33	1598.41	6044.60

As observed, scaling the hypothesis pool M consistently improves grasping success rates with only a negligible sacrifice in inference latency, validating the high efficiency of our parallel generation paradigm. In stark contrast, TTP incurs prohibitive computational overhead and suffers from severe optimization instability. We empirically find that the TTP trajectory frequently diverges from physically viable grasp manifolds, ultimately degrading both execution speed and overall performance.

F Supplementary Visual Results

F.1 OmniDex Benchmark Visual Results

To comprehensively demonstrate the high fidelity and multi-modal richness of our proposed OmniDex benchmark, we present extensive visual galleries for each subset in Figures E1 through E5. For

every representative scene, we visualize the complete suite of aligned multi-modal sensory inputs: multi-view camera images, depth maps, instruction masks, and the physically-validated dexterous grasp ground truths. These visualizations collectively underscore the extreme spatial complexity of the cluttered layouts and the rigorous alignment of our grasping label across diverse visual modalities.

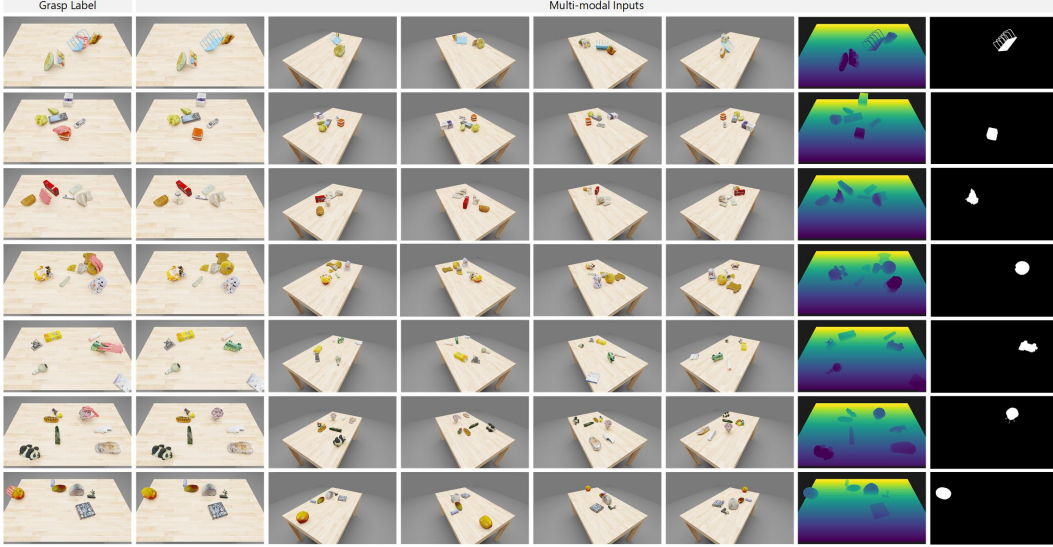


Figure E1: Qualitative visualization of the $\mathcal{D}_{\text{std}}$ subset.

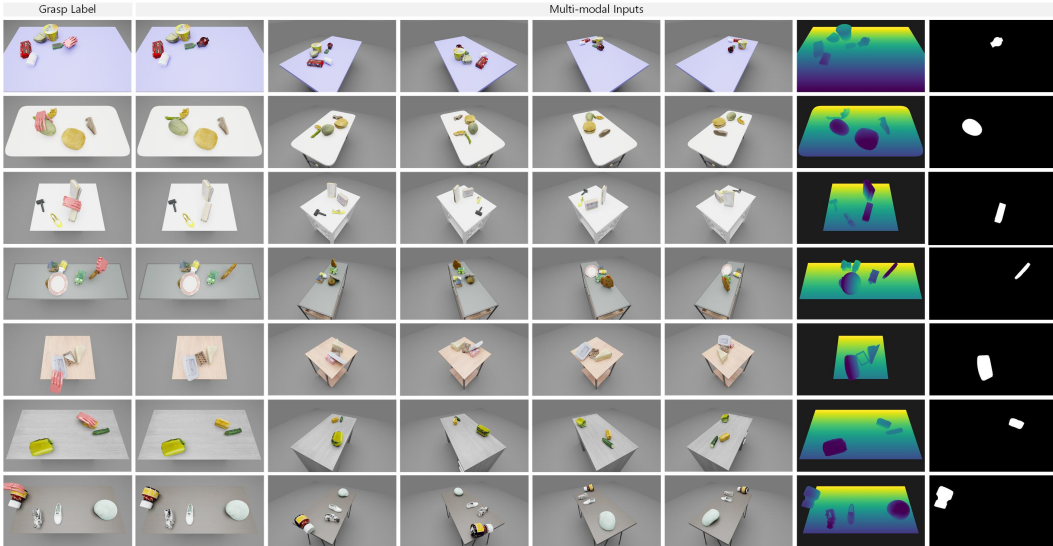


Figure E2: Qualitative visualization of the $\mathcal{D}_{\text{mix}}$ subset.

591 **F.2 OmniDex Model Visual Results**

592 We present qualitative results of OmniDex model across diverse benchmark subsets in Fig. E6. The
 593 predicted grasps remain stable and semantically consistent under substantially different scene layouts.
 594 Notably, OmniDex is able to localize the target object and generate plausible grasping poses even in
 595 the presence of tight inter-object spacing and strong geometric ambiguity among similar instances.
 596 These visualizations further support our quantitative findings, showing that the proposed multimodal
 597 conditioning and physics-aware generation scheme generalize well across varying scene structures
 598 and grasping difficulties.

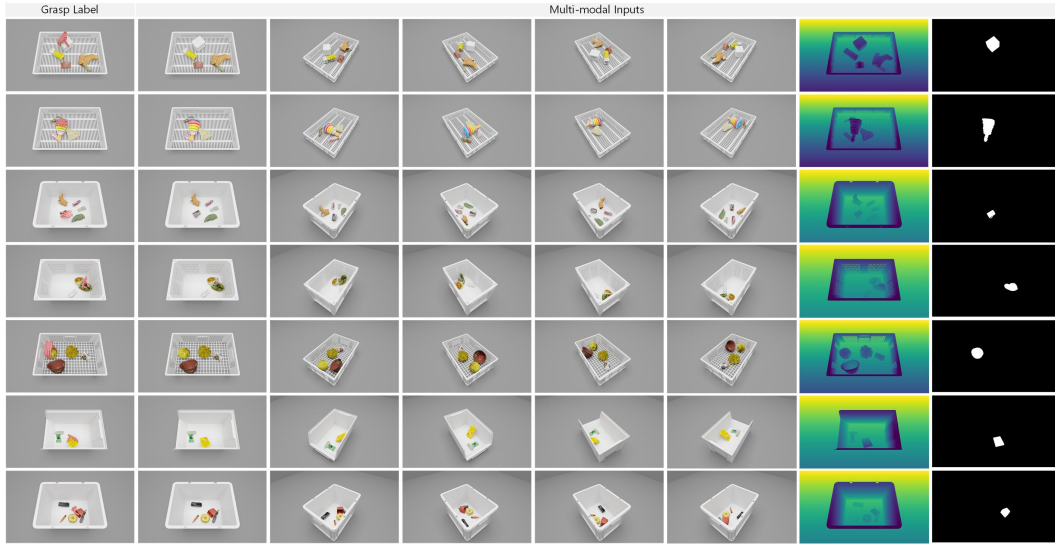


Figure E3: Qualitative visualization of the $\mathcal{D}_{\text{box}}$ subset.

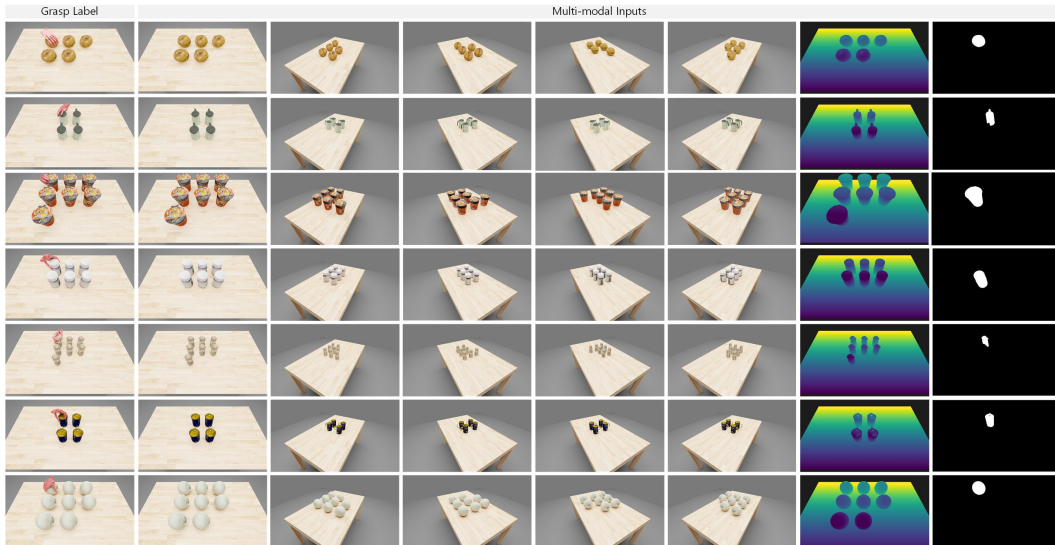


Figure E4: Qualitative visualization of the $\mathcal{D}_{\text{grid}}$ subset.

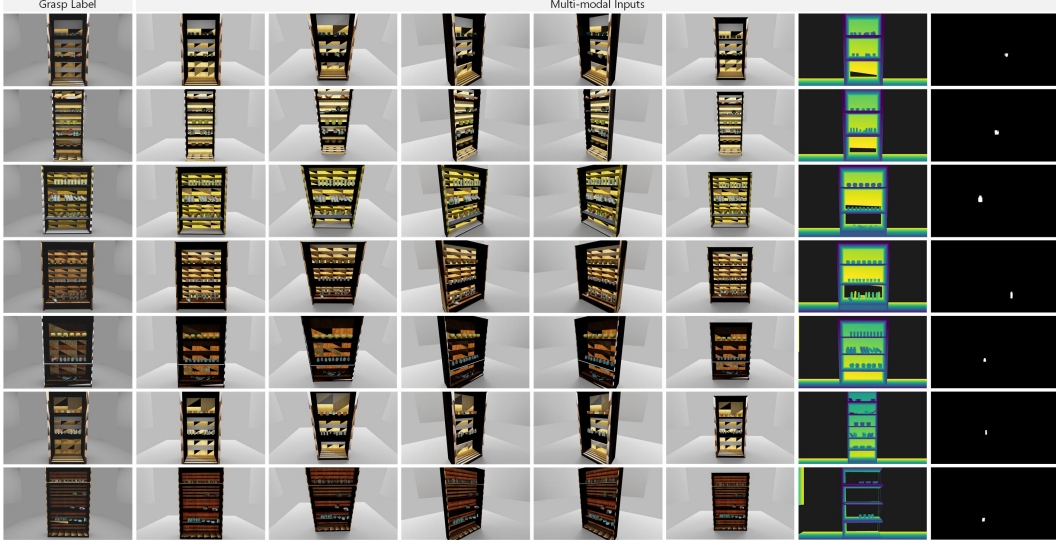


Figure E5: Qualitative visualization of the $\mathcal{D}_{\text{shelf}}$ subset.

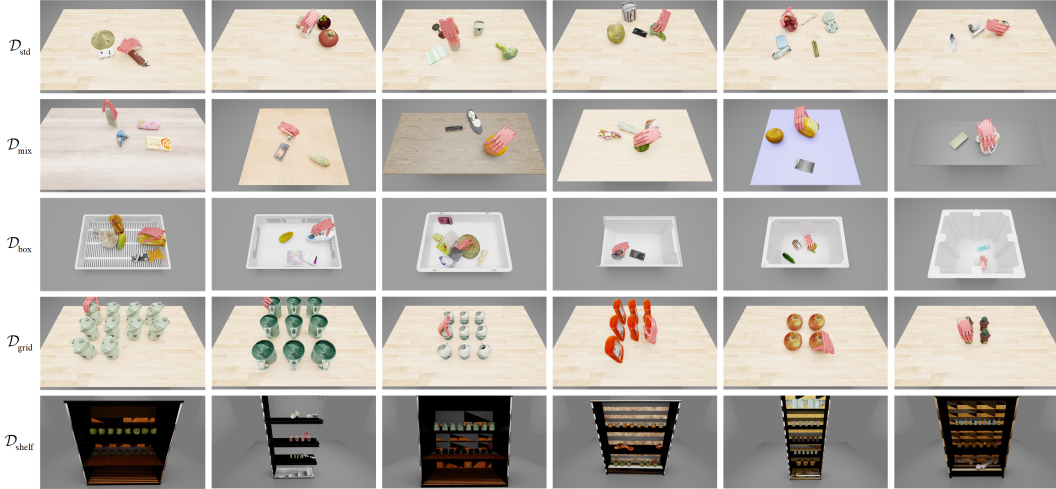


Figure E6: Qualitative results of OmniDex model across diverse benchmark subsets.

599 G Limitation

600 Despite the scale and diversity of the proposed benchmark, both the benchmark construction and
 601 the main evaluation are conducted in simulation. Although we incorporate diverse supporting bases,
 602 multi-view RGB-D observations, and physics-based validation to better approximate realistic cluttered
 603 grasping, a non-negligible sim-to-real gap may still remain due to unmodeled factors such as sensor
 604 noise, calibration error, contact parameter mismatch, and actuation uncertainty. While our seed-and-
 605 filter pipeline is substantially more scalable than scene-level optimization and our inference is more
 606 efficient than test-time optimization, the overall system still requires considerable computational
 607 resources for large-scale data generation and model training, constructing the full benchmark and
 608 training the model involve substantial GPU cost. Addressing real-world transfer and further reducing
 609 the resource requirement are important directions for future work.

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1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction state the main contributions of the paper, namely the CDex benchmark, the CDex model, and the empirical gains in cluttered-scene dexterous grasping. These claims are supported by the benchmark description in Sections 3 and Appendix A–B, the method in Section 4 and Appendix C, and the experiments in Section 5 and Appendix D–E.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: The paper discusses limitations in the supplementary material, including the simulation-to-real gap and the computational cost of large-scale benchmark construction and model training; see the final limitations section in the appendix.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: The paper describes the benchmark construction pipeline, model architecture, loss functions, evaluation protocol, and dataset splits in Sections 3–5 and Appendix A–D. These details are sufficient to understand and reproduce the main reported results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [\[Yes\]](#)

Justification: We will provide anonymized access to the code and released benchmark data at submission time, together with documentation and instructions for reproducing the main experimental results, as described in the supplementary material.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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Justification: The paper specifies the train/test splits, benchmark subsets, optimizer, training schedule, loss weights, and inference settings in Sections 3–5 and Appendix B–D. Additional implementation details, including compute cost and parallel hypothesis settings, are provided in Appendix D.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

659 Answer: [No]

660 Justification: The paper reports point estimates for all quantitative results, but it does not

661 include error bars, confidence intervals, or formal statistical significance tests.

662 **8. Experiments compute resources**

663 Question: For each experiment, does the paper provide sufficient information on the com-

664 puter resources (type of compute workers, memory, time of execution) needed to reproduce

665 the experiments?

666 Answer: [Yes]

667 Justification: The supplementary material reports the compute worker type (NVIDIA A800

668 GPUs), the total compute used for benchmark generation, the training configuration, and the

669 inference latency.

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679 societal impacts of the work performed?

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681 Justification: The current submission does not include a dedicated broader impacts discus-

682 sion. While the work is motivated by embodied AI and robotics applications, potential

683 beneficial and adverse downstream impacts are not explicitly analyzed in the paper.

684 **11. Safeguards**

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687 image generators, or scraped datasets)?

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689 Justification: The paper does not describe the release of a high-risk generative model or

690 scraped internet-scale dataset of the type targeted by this question. The presented work

691 focuses on simulation-based dexterous grasping benchmarks and models.

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696 Answer: [Yes]

697 Justification: The paper credits all third-party assets used to construct the benchmark, and

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700 **13. New assets**

701 Question: Are new assets introduced in the paper well documented and is the documentation

702 provided alongside the assets?

703 Answer: [Yes]

704 Justification: The paper introduces the CDex benchmark as a new released asset. The

705 supplementary material describes its construction pipeline, data splits, usage scope, and

706 release documentation provided with the dataset.

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